SKOG SHADER GUIDE

Quick Start

1. Create a new Material and assign the **SkogShader**, **SkogBillboardShader**, or **SkogGrassShader**.
2. Assign **BaseColor**, **Normal**, **AOMap**, and **SplatMap**.
3. Adjust **LeafColor**, **BarkColor**, and **Moss Color** to your liking.
4. Enable **Wind**, **Random Tinting**, or **Slope Effects** if you wish.

Performance Tips

All provided materials have GPUInstancing ticked. If you would like to use static batching on your prefabs, please disable this feature from the materials you use.

Texture and Model Requirements

If you wish to use these shaders on your own custom models, there are several important requirements for your textures.

1. **BaseColor Map** - This must be a greyscale image, or splat map tinting may not work as expected. It is recommended to import with **Alpha Is Transparency** checked and **sRGB (Color Texture)** disabled. Enabling **Preserve Coverage** is also recommended for better transparency at far distances.
2. **AO Map** - It is recommended to keep this texture soft and blurry. This texture's imported **Max Size** can be lowered significantly e.g. 256 and it is recommended to disable **sRGB (Color Texture)**.
3. **Roughness Map** - This texture's **Max Size** can be lowered to half.
4. **Normal Map** - This texture should be marked as a **Normal Map** on import.
5. **Splat Map** - This texture should contain solid areas of red, green, and blue, masking out the sections of your model you would like to tint separately. Red is generally for tinting leaves. Green is generally for tinting bark. Blue is generally for tinting moss or other extras.
6. **Wind** - Wind is driven by vertex colors on your model. Make sure areas you would like to be moved by wind, have a white vertex color.

Shader Feature Lookup

1. SkogShader

A custom shader graph designed for foliage and rocks and includes features such as wind, splat maps, random tinting, and slope effects for snow or moss.

- **BaseColor** - Albedo map.
- **Alpha Threshold** - Alpha Clip threshold.
- **Alpha Softness** - How soft the alpha fading will be.
- **Normal** - Normal map.
- **Normal Strength** - How strong the normal map will be.
- **Roughness** - Roughness map.
- **Smoothness** - Adjusts the strength of the roughness map, or, if there is no roughness map assigned, will set the smoothness directly.
- **AOMap** - Ambient Occlusion map (dark shaded parts of the material).
- **AOStrength** - The strength of the Occlusion Map.

Translucency

- **Translucency** - Whether translucency should be enabled on the shader.
- **TranslucencyColor** - The color of the translucency effect.
- **TranslucencyStrength** - Strength of the Translucency effect.

Tinting

- **SplatMap** - An rgb map designating where custom tints on the material should be assigned. e.g. Red sections tint leaves. Green sections tint bark. Blue sections tint moss or other extras.
- **LeafColor** - Color of the leaves. If no splat map is assigned this will tint the whole material.
- **BarkColor** - Color of the Bark.
- **MossColor** - Color of the moss or other extras.
- **RandomTinting** - Whether random tinting based off of a Simple Noise texture should be used.
- **RandomTintScale** - The scale of the Simple Noise texture.
- **RandomTintColor** - The tint applied via the Simple Noise texture.

Wind

- **Wind** - Whether wind should be applied.
- **WindTexture** - A grayscale map that is scrolled to displace the foliage vertices, producing a wind effect.
- **WindSpeed** - The x,z speed and scroll direction of the wind.
- **WindStrength** - The strength of the wind effect.

SlopeEffect

- **SlopeEffect** - Whether a slope effect should be used. This is useful for assigning snow in reasonable places.
- **SlopeTexture** - The texture that will be assigned in the slope. e.g. snow, moss.
- **SlopeHeight** - How high in the up direction the slope effect will be.
- **SlopeBlend** - How sharp the slope edge will be.

2. SkogBillboardShader

A custom shader graph designed for rotating a card around the y axis to face the camera. It also includes splat maps and random tinting.

- **BaseColor** - Albedo map.
- **Clip** - Alpha Clip threshold.
- **Normal** - Normal map.
- **NormalStrength** - How strong the normal map will be.

<u>Tinting</u>

- **SplatMap** - An rgb map designating where custom tints on the material should be assigned. e.g. Red sections tint leaves. Green sections tint bark. Blue sections tint moss or other extras.
- **LeafColor** - Color of the leaves. If no splat map is assigned this will tint the whole material.
- **BarkColor** - Color of the Bark.
- **MossColor** - Color of the moss or other extras.
- **RandomTinting** - Whether random tinting based off of a Simple Noise texture should be used.
- **RandomTintScale** - The scale of the Simple Noise texture.
- **RandomTintColor** - The tint applied via the Simple Noise texture.

3. SkogGrassShader

A custom shader graph designed for coloring grass or other small props based on the y coordinate of the uv map. Top and bottom colors blend vertically based on the V coordinate (UV.y). The shader includes wind and random tinting.

- **TopColor** - The top color of the grass.
- BottomColor - The bottom color of the grass.

<u>Tinting</u>

- **RandomTinting** - Whether random tinting based off of a Simple Noise texture should be used.
- **RandomTintScale** - The scale of the Simple Noise texture.

- **RandomTintColor** - The tint applied via the Simple Noise texture.

__Wind__

- **Wind** - Whether wind should be applied.
- **WindTexture** - A grayscale map that is scrolled to displace the foliage vertices, producing a wind effect.
- **WindSpeed** - The x,z speed and scroll direction of the wind.
- **WindStrength** - The strength of the wind effect.

Troubleshooting

- ***Foliage looks too bright or dark***

Check if your material has translucency or random tinting enabled. Disable these features temporarily to see if it fixes the issue.

- ***Wind isn't working***

Make sure you have assigned a grayscale wind texture to your material, and your model has non-black vertex colors. Also make sure you have set a non-zero wind speed and wind strength.

Version 1.0 - 2025
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